NOVEMBER 23, 2015

MATH 103–113 MIDTERM 2 SOLUTION KEY

FALL 2015

1. (8p) Let $f(u) = \left(\frac{u-2}{u+1}\right)^2$ and $u = \tan x$. Find $\frac{df}{dx}$ at $x = 0$.

$$\frac{df}{dx} = \frac{df}{du}\frac{du}{dx} \quad \text{(Chain Rule)}$$

For $x=0$, $u = \tan 0 = 0$

1p

$$= 2\left(\frac{u-2}{u+1}\right)\left[\frac{(\cancel{u}+1)-(\cancel{u}-2)}{(u+1)^2}\right]\sec^2 x$$

5p

$$\left.\frac{df}{dx}\right|_{x=0} = \left.\frac{+6(u-2)}{(u+1)^3}\right|_0 \left.\frac{1}{\cos^2 x}\right|_0 = \frac{-12}{1}\cdot 1 = -12$$

2p

2. (10p) The function $y(x)$ is determined *implicitly* by the equation $(x-1)^y = \left(\frac{y}{2}-1\right)^x$. Find an equation for the line *tangent* to the curve $y(x)$ at the point $(2,4)$.

Take the natural logarithm of both sides

2p $\quad y\ln(x-1) = x\ln\left(\frac{y}{2}-1\right) \rightsquigarrow y = y(x)$ (implicit f.)

$$\frac{dy}{dx}\ln(x-1) + \frac{y}{x-1} = \ln\left(\frac{y}{2}-1\right) + \frac{\frac{1}{2}x\frac{dy}{dx}}{\frac{y}{2}-1}$$

4p $\quad \frac{dy}{dx}\left[\ln(x-1) - \frac{x}{y-2}\right] = \ln\left(\frac{y}{2}-1\right) - \frac{y}{x-1}$

$$\left.\frac{dy}{dx}\right|_0 = \frac{\ln 1 - \frac{4}{1}}{\ln 1 - \frac{2}{2}} = \frac{4}{1} = 4$$

2p

Eq. for the TL through $(2,4)$ with slope $m = 4$

$$y - 4 = 4(x-2) \rightarrow y = 4x - 4$$

2p

3. (8p) Find the *linearization* of $f(x) = \arctan(2x)$ at the point $a = 0$. Use the linearization to approximate $\arctan\left(\frac{1}{12}\right)$.

$f(x) \approx L(x) = f(0) + f'(0)(x-0)$ for x near 0

$f(0) = \arctan 0 = 0$; $f'(x) = \frac{2}{1+(2x)^2}$ $f'(0) = 2$

4p

$f(x) \approx L(x) = 2x$ 2p

For $x = \frac{1}{24}$ we have $\arctan\left(\frac{1}{12}\right) \approx \frac{1}{12}$ 2p

4. (6p) The radius r of a sphere decreases from $a = 1$ m to 0.99 m. Use *differential* dV to estimate the *decrease* in the sphere's volume $V(r) = \frac{4\pi r^3}{3}$.

$r_0 = a = 1$ $dr = 0{,}99 - 1$

$V'(r) = 4\pi r^2$ $dr = -0{,}01$ m 1p

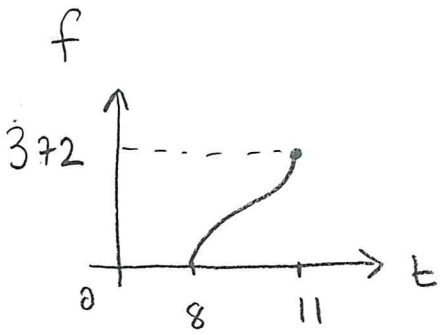
$dV = V'(r_0)\,dr = 4\pi r_0^2\,dr$ 3p

An estimate for the decrease in the volume

$dV = 4\pi(1)^2(-0{,}01)$

$= -0{,}04\pi$ 2p

5. (6p) Anatolia Motorway is about 372 km long and has a speed limit of 120 km/h. A car enters the motorway at 8 AM and reaches the end at 11 AM. As the car exits, the driver is stopped and given a speeding ticket. Use the *Mean Value Theorem* for derivatives to explain this situation.



MVT: $f'(c) = \dfrac{f(b)-f(a)}{b-a}$, $a<c<b$

2p

$f'(c) = \dfrac{372-0}{11-8} = 124 > 120$ 4p

speed at c

speed limit

6. (12p) Find the *absolute maximum and minimum values* of $f(x) = \sqrt[3]{x}(28 - x^2)$ on the interval $[-1, 8]$.

f is continuous on $[-1,8]$. Thus it assumes its absolute extreme values on $[-1,8]$.

CP(s): $f'(x) = \frac{1}{3}x^{-\frac{2}{3}}(28-x^2) + x^{\frac{1}{3}}(-2x) = \dfrac{7(4-x^2)}{3x^{2/3}}$

$f'=0$ at $\cancel{x=-2}$ and $x=2$; f' is not defined at $x=0$

outside the interval

6p

CPs values : $f(0)=0$ $\quad f(2) = 2^{\frac{1}{3}}24$ abs. max value

End point values: $f(-1) = -27$ $\quad f(8) = 2(-36)$

abs min. value.

6p

7. (10p) Let $0 < x < 20$ and $0 < y < 20$. The sum of x and y is 20. Find x and y so that $x\sqrt{y}$ is to be as large as possible. (Hint: Use the *First Derivative Test*).

$x+y=20 \rightarrow y=20-x$; $f(x)=xy=x\sqrt{20-x}$ 3p

CP: $f'(x)=\sqrt{20-x}-\dfrac{x}{2\sqrt{20-x}}=\dfrac{2(20-x)-x}{2\sqrt{20-x}}=0$

$\rightarrow 40-3x=0 \rightarrow x=\dfrac{40}{3}$ 3p

x	0		$40/3$		20
f'		$-$	0	$+$	
f		$\nearrow$	lmax	$\searrow$	

By the First derivative test

f has a local (also abs.) max. value at $x=\dfrac{40}{3}$

Conc: $x=\dfrac{40}{3}$ $y=20-x=\dfrac{20}{3}$ 4p

8. (15p) Find the limits using the *L'Hopital Rule.*

a) $\lim\limits_{t\to 0}\dfrac{t\sin t}{1-\cos t}=\left(\dfrac{0}{0}\right)=\lim\limits_{t\to 0}\dfrac{\sin t+t\cos t}{\sin t}\left(\dfrac{0}{0}\right)$

$=\lim\limits_{t\to 0}\dfrac{2\cos t+t\sin t}{\cos t}=\dfrac{2}{1}=2$ 6p

b) $\lim\limits_{x\to 0}\overbrace{(e^x+x)^{\frac{1}{x}}}^{f(x)}=$ $(1^\infty$ indeterminate power$)$

$\ln f(x)=\dfrac{1}{x}\ln(e^x+x)$

$\lim\limits_{x\to 0}\ln f(x)=\lim\limits_{x\to 0}\dfrac{\ln(e^x+x)}{x}\left(\dfrac{0}{0}\right)=\lim\limits_{x\to 0}\dfrac{\frac{e^x+1}{e^x+x}}{1}=2$

Thus $\lim\limits_{x\to 0}f(x)=e^2$ 9p

9. (25p) Let $f(x) = \dfrac{\ln x}{x}$. *a)* Identify the domain of f. $x > 0$ or $(0, \infty)$ 1p

6p *b)* Find the critical point(s) of f, if any, and determine where the local extreme value(s) of f occur. Find where the curve is increasing and where it is decreasing.

$$f'(x) = \frac{\frac{1}{x}x - \ln x}{x^2} = \frac{1 - \ln x}{x^2} = 0 \xrightarrow{CP} x = e$$

x	0		e		∞
sign of f'		$+$	0	$-$	
f		$\nearrow$	l. max	$\searrow$	

f is $\nearrow$ on $(0, e)$

f is $\searrow$ on (e, ∞)

6p *c)* Determine the intervals where the curve is concave up and where it is concave down. Identify the inflection point(s), if any.

$$f''(x) = \frac{-\frac{1}{x}x^2 - 2x(1 - \ln x)}{x^4} = \frac{2\ln x - 3}{x^3} = 0 \rightarrow x = e^{\frac{3}{2}} \text{ possible IP}$$

> 0 for $x > 0$

x	0		$e^{3/2}$		∞
sign of f''		$-$	0	$+$	
f		$\frown$		$\smile$	

f is concave up on $(e^{3/2}, \infty)$

f is concave down on $(0, e^{3/2})$

2p *d)* Find the *horizontal asymptote* of the graph of f by computing $\lim_{x\to\infty} f(x)$.

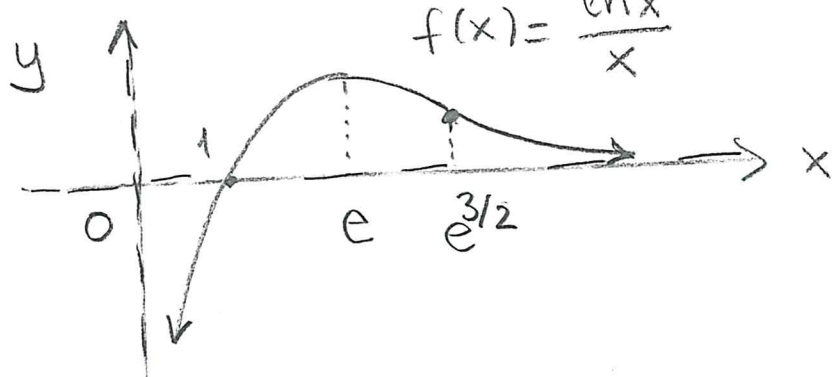
$$\lim_{x\to\infty} \frac{\ln x}{x} \left(\frac{\infty}{\infty}\right) = \lim_{x\to\infty} \frac{\frac{1}{x}}{1} = 0 \rightsquigarrow y = 0 \text{ is a HA}$$

2p *e)* Find the *vertical asymptote* of the graph of f, and compute $\lim_{x\to 0^+} f(x)$.

VA at $x = 0$. $\displaystyle\lim_{x\to 0^+} \frac{\ln x}{x} = \lim_{x\to 0^+} \frac{1}{x} \cdot \ln x = (+\infty)(-\infty) = -\infty$

8p *f)* Use the above information to sketch the graph of the curve.

x	0		1		e		$e^{3/2}$		∞
f	$-\infty$	$\nearrow$	0	$\nearrow$	l. max	$\searrow$	IP	$\searrow$	0
		$\frown$		$\frown$		$\frown$		$\smile$	



HA $y = 0$

VA, $x = 0$